

Week 01 — Before You Come

100L • Semester 1 • How Computers & the Web Work

The first session answers one question: **what actually happens when you open a website?** You will be using the Web for the next four years as builders, not just users. This page sets you up for the first session, and it takes about 45 minutes. Do the three tasks in order. The order matters.

Task 1: The "Today" log (10 min, no data needed)

Pick up your phone and look back over the last day. Write down **eight things you did with it**. Include at least one money transfer, one phone call, and one message.

Then copy this table and tick one column for each:

What I did	Used the Web	Used the Internet, but not the Web	Used neither	Not sure
e.g. checked my result on the school portal				

Don't look anything up; guess. One hint: one of the most common ways people in Nigeria send money does not use the Internet at all. We settle every answer together in the first ten minutes of the session.

Task 2: The "Before" sketch (5 min, do this *before* reading)

On one page, draw what you think happens between typing a web address into your browser and the page appearing. Use boxes and arrows, and label whatever you can.

It is meant to be incomplete and probably wrong. Nothing is marked. At the end of the session you will draw it again, and comparing the two drawings is how you will see what you learned.

Task 3: Read (about 30 min)

Read these in order. Each one has a question to keep in mind while you read. You don't need to write the answers down; the session uses them.

1. **MDN: How the web works** (15 min) https://developer.mozilla.org/en-US/docs/Learn_web_development/Getting_started/Web_standards/How_the_web_works

Keep in mind: the article compares a website to a shop. What does each part of the comparison stand for: the house, the road, the shop, the language you order in?

2. **Flavio Copes, HTTP course: "What HTTP does" and "Clients and servers"** (5 min each)
<https://flaviocopes.com/courses/http/what-http-does/>
<https://flaviocopes.com/courses/http/clients-and-servers/>

Keep in mind: can one program be both a client and a server? When? (Both lessons show a `curl` command to type into a terminal. Skip it for now; the terminal is week 4.)

3. The Valley of Code: DNS (3 min) <https://flaviocopes.com/dns/>

Keep in mind: why do we type names like `google.com` instead of the numbers computers actually use?

Optional, if you want more:

- [MDN: The web standards model](#): who invented the Web, and why no company owns it.
- [MDN: How browsers load websites](#): what your browser does with the files once they arrive.

No data this week? The readings are optional for you. Tasks 1 and 2 matter more, and they need no connection.

Bring with you

- Your **Today log** and your **Before sketch**
- A notebook and a pen. This week is spent on paper and on your feet, not on a screen.
- A phone, if you have one
- A laptop is **not** needed this week. Tools are installed in week 4. If you plan to buy or borrow one, check it has at least 4 GB RAM and 20 GB free space now. 8 GB RAM and 50 GB free is comfortable.

How this programme runs

- **One session a week, about 3½ hours.** It runs in five parts: a short warm-up, a live concept segment, a long build (lab) segment, a short show-and-tell, and a submission at the end. Most of the time goes to building, which is why reading before you come matters.
- **Missing a week is recoverable.** The class code repository is tagged at the start of every week, and you can pick up from that point.
- **Your work history counts.** From week 11 you will keep your work in Git, which records every change you make. A project with no believable history of work across the semester does not pass, however good it looks at the end. This rule holds until your final-year project.